

SMART JOB PORTAL SYSTEM

with AI-Powered Recruitment Assistance

An intelligent web-based application designed to enhance recruitment through AI integration and modern software architecture.

1. Abstract

The Smart Job Portal System is developed using ASP.NET Core Web API and Entity Framework Core, following Clean Architecture principles for scalability and maintainability. It connects job seekers and recruiters through a centralized platform that supports job searching, applications, and hiring processes. Key features include smart job recommendations, match score calculation, skill gap analysis, and an AI chatbot for real-time assistance.

2. Problem Statement

Many existing platforms lack personalization and efficient job matching, relying on basic keyword filtering that results in irrelevant job recommendations. Candidates receive little guidance on skill improvement, while recruiters face challenges in identifying suitable candidates quickly among a large pool of applicants.

3. Objectives

- Build an intelligent and scalable recruitment platform.
- Integrate AI features like job recommendations, match scoring, and skill gap analysis.
- Ensure high performance and security using JWT authentication and role-based access control.
- Provide real-time assistance through an AI chatbot.

4. System Architecture

The system is built using Clean Architecture to ensure separation of concerns across four distinct layers:

Layer	Responsibility
Domain	Core entities and business rules.
Application	Business logic and AI feature implementation.
Infrastructure	Database management (SQL Server) and external services.
API	Handling client requests and RESTful endpoints.

5. AI Module & Features

The AI module transforms the platform into an interactive assistant:

- **Job Recommendations:** Based on user skills and profiles.
- **Match Score:** Measures suitability for specific job roles.
- **Skill Gap Analysis:** Identifies missing skills to guide career growth.
- **Chatbot:** Provides real-time support and improves user engagement.

6. User Roles

The system supports three user roles with specific access levels:

- **Admin:** Oversees operations, user control, and platform integrity.
- **Recruiter:** Manages job postings and reviews applications.
- **Candidate:** Manages profiles, uploads resumes, and applies for jobs.

7. Future Scope

Planned enhancements include the development of a mobile application, integration of video interview features, advanced machine learning models, and connecting with online learning platforms for global job market support.

